For Example
int num=12;

num1 variable is declared as integer types. So, it can store integer value

(2) char data type:

Character data type accepts and returns all valid ASCII characters. Character data type is often said to be an integer type, since all the characters are represented in memory by their associated **ASCII Codes.** If a variable is declared as char, C++ allows storing either a character or an integer value.

Example 1:-

char c='A';

cout<<ch ;

In the above code, **ch** is declared as a char type variable to hold a character. It displays the character A

Example 2:-

char ch='A'

cout<<ch+1;

In the above statements, the value of **ch** is incremented by 1 and the new value is stored back in the same variable **ch**. (Remember that, arithmetic operations are carried out only on the numbers not with alphabets) so it displays B

Another program illustrates how **int** and **char** data types are working together.

Illustration 9.3: C++ program to get an ASCII value and display the corresponding character

```
#include <iostream>
using namespace std;
int main ()
{
        int n;
        char ch;
        cout << "\n Enter an ASCII code (0 to 255): ";
        cin >> n;
        ch = n;
        cout << "\n Equivalent Character: " << ch;
}
```

The output

```
Enter an ASCII code (0 to 255): 100
Equivalent Character: d
```

In the above program, variable **n** is declared as an int type and another variable **ch** as a char type. During execution, the program prompts the user to enter an ASCII value. If the user enters an ASCII value as an integer, it will be stored in the variable **n**. In the statement **ch = n**; the value of **n** is assigned into **ch**. Remember that, **ch** is a char type variable.

Chapter 9 Page 115-151.indd 134 3/24/2020 9:21:05 AM